

Bayesian Statistics

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1 Posterior Distribution

θ is the parameter. D is data. $P(\theta)$ is prior distribution. $P(\theta|D)$ is posterior distribution.

Probability of parameter and data is defined as: $P(\theta \cap D) = P(\theta)P(D|\theta) = P(D)P(\theta|D)$

Posterior distribution can be calculated by rearrangement: $P(\theta|D) = \frac{P(\theta)P(D|\theta)}{P(D)}$

2 Markov chain Monte Carlo

MCMC estimates the posterior, $P(\theta|D)$, without estimating $P(D)$.

The goal of MCMC is to find a range with the parameter.

MCMC process:

1. Choose a starting point in parameter space based on the prior .
2. Move from one point to another based on: $P(\frac{A \rightarrow B}{B \rightarrow A}) \propto \frac{Post(B)}{Post(A)}$, where A and B are two points in parameter space.
3. The process will spend more time around a certain point.
4. The process stops after the posterior distribution is not changing (converges).

5. The program will output a list of all points explored during the process.
6. A range with the true parameter is be picked from the list.